

Microteaching Session

From Classical Bits to Quantum Bits

Dr. Sreekar Guddeti

Learning Outcomes

- Explain a classical bit.
- Describe a qubit.
- Understand superposition qualitatively.
- Mention one application of quantum computing.

Motivation

Imagine searching for a friend in a city with one million houses.

Motivation

Imagine searching for a friend in a city with one million houses.
A classical computer checks one possibility after another.

Motivation

Imagine searching for a friend in a city with one million houses.
A classical computer checks one possibility after another.
Can Nature help us compute differently?

Motivation

Imagine searching for a friend in a city with one million houses.
A classical computer checks one possibility after another.
Can Nature help us compute differently?

Today's Question

What makes a quantum computer different?

Classical Bit

A classical bit has only two states:

0 or 1

Classical Bit

A classical bit has only two states:

0 or 1

Examples:

- Light switch: ON/OFF
- Memory cell
- Voltage High/Low

Classical Bit

A classical bit has only two states:

0 or 1

Examples:

- Light switch: ON/OFF
- Memory cell
- Voltage High/Low

Question: Can a classical bit be both 0 and 1 simultaneously?

Quantum Bit (Qubit)

A qubit is the basic unit of quantum information.

Quantum Bit (Qubit)

A qubit is the basic unit of quantum information.

Unlike a classical bit, a qubit can exist in a **superposition** of states.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Quantum Bit (Qubit)

A qubit is the basic unit of quantum information.

Unlike a classical bit, a qubit can exist in a **superposition** of states.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Measurement gives either 0 or 1.

Superposition Analogy

Classical Coin

- Heads
- Tails

Spinning Coin

- Appears to show both possibilities
- Only an analogy

Superposition Analogy

Classical Coin

- Heads
- Tails

A qubit is fundamentally a quantum object—not literally a spinning coin.

Spinning Coin

- Appears to show both possibilities
- Only an analogy

Why Quantum Computing?

Potential applications:

- Cryptography
- Drug discovery
- Material science
- Optimization
- Quantum simulation

Why Quantum Computing?

Potential applications:

- Cryptography
- Drug discovery
- Material science
- Optimization
- Quantum simulation

Important: Quantum computers are *not* faster for every problem.

Think–Pair–Share

Two classical bits produce:

00, 01, 10, 11

Think–Pair–Share

Two classical bits produce:

00, 01, 10, 11

Question:

What changes if we replace them with two qubits?

Discuss with your neighbour for one minute.

Summary

- Classical bit: 0 or 1.
- Qubit: quantum state.
- Superposition is a uniquely quantum property.
- Quantum computing enables new algorithms for selected problems.

Exit Question:

Name one application of quantum computing.

Thank You

Questions?